

# Visual Product Similarity Search

Conversation deck for explaining the demo options while the app is active

AI-powered product image matching for packaging, catalog, SKU, and QA teams.

Use this as a conversation guide, not a formal long presentation.

Main demo remains the mobile/static or live app.

## 30-second pitch

I built a visual product similarity system for companies with many product images or packaging versions.

It helps teams find duplicates, variants, wrong images, and similar packaging faster.

## Presenter

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Full-stack Developer and Security Researcher

# The Business Problem

Companies manage many product images, SKUs, barcodes, sizes, flavors, and packaging versions.

Similar packages can create duplicate catalog records or QA confusion.

Manual visual checking becomes slow when the catalog grows.

Barcode data alone does not solve visual packaging comparison.

## Simple example

Two products may have different barcodes but very similar packaging.

The team needs to know: duplicate, variant, old packaging, or catalog mistake?

## Who feels this pain?

Catalog teams

SKU/data teams

QA teams

Packaging teams

Retail/distribution teams

# The Solution

Upload, select, or capture a product image.

The system converts the image into an AI embedding.

It searches the product catalog using vector similarity.

It returns the closest matching products with scores.

The team reviews duplicates, variants, or possible mistakes faster.

## Flow

Product image

AI image understanding

Vector search

Top similar products

## Business value

Less manual checking

Cleaner catalog data

Faster packaging comparison

Better QA review

# Demo Options You Can Show

## Static mobile demo

- Fast and reliable at the event.
- Opens on phone browser.
- Best when internet is weak.
- Shows the user flow and business idea.

## Live backend demo

- Stronger technical proof.
- Uses deployed app/backend.
- Shows real search behavior.
- Requires stable internet and running server.

## Screenshots/video

- Final fallback.
- Useful for quick booth conversations.
- Protects you if venue network is bad.
- Good for follow-up messages.

# Product Options for Customers

## Android app

Field use and quick checks.  
Camera/gallery image capture.  
Top similar products on phone.  
Best for sales, warehouse, QA, and field staff.

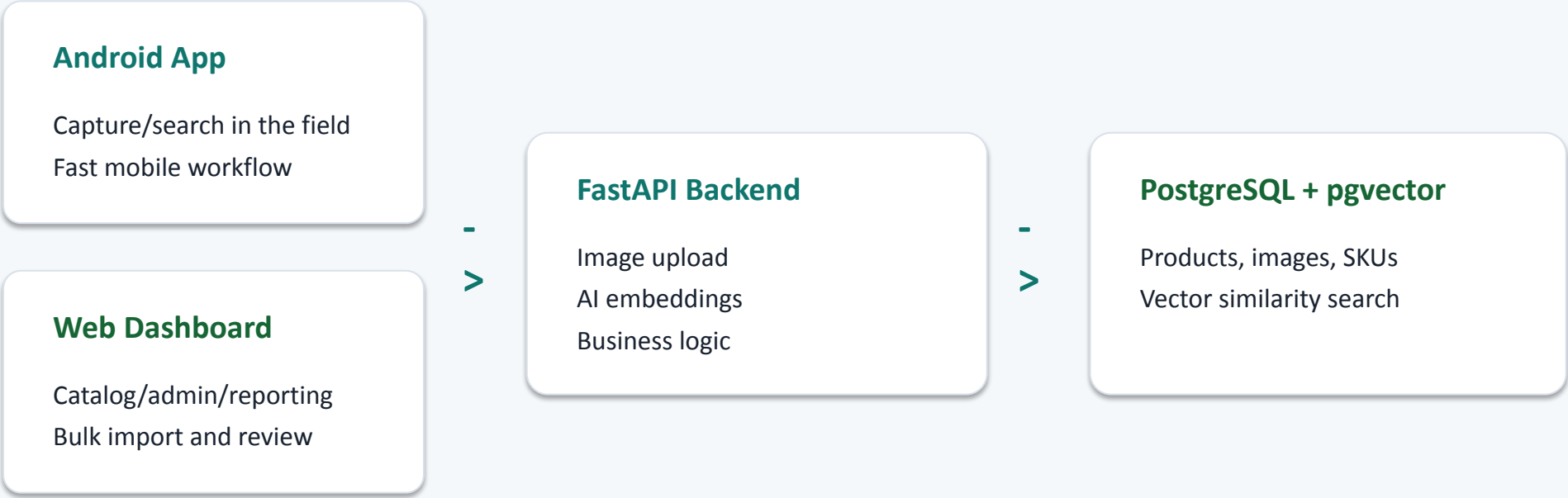
## Web dashboard

Catalog and admin work.  
Bulk upload products.  
Review duplicates and variants.  
Export reports for teams.

## API integration

Connect to ERP, ecommerce, WMS, or inventory systems.  
Keeps existing workflow.  
Adds visual search as a service.  
Best for mature companies.

# Recommended Commercial Architecture



## Important

Streamlit is useful for prototype/demo work.  
It should not be the final commercial laptop product.

# Where It Fits at Fi Africa / ProPak

## Packaging and labeling

Compare packaging designs.  
Detect similar or duplicate packaging.  
Support QA and client review.

## Food manufacturers

Manage product variants, flavors, and sizes.  
Compare old vs new packaging.  
Keep product images consistent.

## Distributors and retailers

Clean catalog duplicates.  
Verify product images across suppliers.  
Improve product search.

## Automation / QA companies

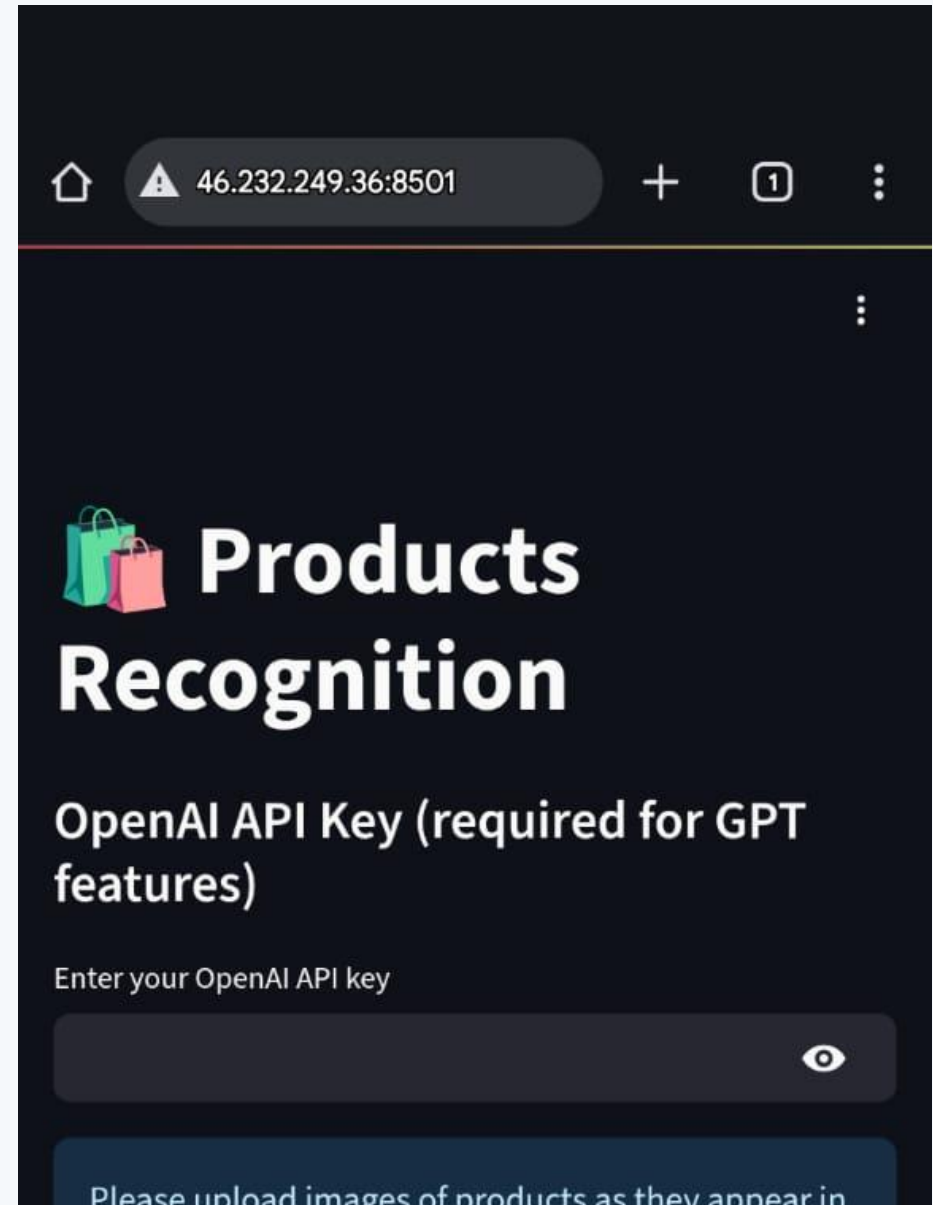
Add visual similarity to inspection or traceability workflows.  
Strong ProPak MENA fit.

## Best first target

Hall 1 / ProPak: packaging, labeling, coding, QA, automation.

# Demo Flow

1. Choose or show a reference product.
2. Click “Find Similar Products” or upload/capture an image.
3. Show the top 3 matches.
4. Explain similarity score: higher means visually closer.
5. Connect to value: duplicate, variant, wrong image, or packaging review.





# Questions to Ask Customers

Do you manage many SKUs or product images?

Do you have duplicate or similar packaging problems?

Do your teams compare old vs new packaging versions?

Do you need to verify product images against barcode/SKU records?

Do you use ERP, ecommerce, WMS, or inventory software?

Who owns product catalog quality in your company?

## What you are qualifying

Need for demo

Need for pilot

Need for API integration

Right person for follow-up

## Ask for next step

Technical meeting

Sample product images

Pilot discussion

Referral to catalog/QA/IT owner

# Suggested Pilot

## Pilot idea

Test the system with 100-500 real product images from the customer.

## Customer provides

Product images  
Product names  
SKU/article numbers  
Barcodes if available  
Examples they want to detect

## Pilot deliverables

Catalog import  
Image similarity search  
Top 3/top 5 matches  
Duplicate/variant review  
Short findings report

## Success criteria

Finds useful similar products.  
Reduces manual review time.  
Identifies confusing product records.  
Shows clear value for catalog or QA workflow.

# Technical Foundation

Use only if the customer is technical

Python / FastAPI backend.

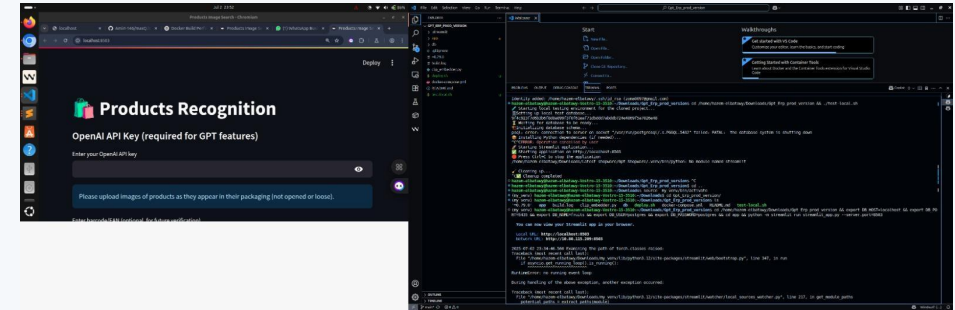
CLIP image embeddings.

PostgreSQL with pgvector for vector search.

Android app for mobile capture/search.

Web dashboard for catalog/admin workflow.

API integration possible with ERP, inventory, and ecommerce systems.



# Wording That Works

## Use these phrases

Visual product similarity search  
Product catalog quality  
Duplicate and variant detection  
Packaging comparison  
SKU and barcode data support  
Mobile capture/search  
Web dashboard for catalog management

## Avoid these claims

This is only a demo  
It replaces your ERP  
It guarantees 100% accuracy  
It works perfectly on every product image

## Best closing line

It helps your team review similar products faster and reduce manual checking.

# What Makes Product Similarity Different?

Answer when someone says: we already have barcode, ERP, or normal product search

## Visual search beyond barcode

Barcode tells you identity.  
This system compares how products actually look: packaging, label, shape, and visual similarity.

## AI embeddings, not filename matching

Uses CLIP/OpenCLIP image embeddings.  
Compares image meaning/features instead of simple text or manual tags.

## Vector database search

Stores image embeddings in PostgreSQL + pgvector.  
Designed for fast top-match search as the catalog grows.

## Search before upload

Find similar products already in the database.  
Useful for review and discovery without adding new records.

## Duplicate prevention

When uploading a new product, the system can warn about visually similar existing products.  
Helps avoid duplicate catalog entries.

## Business workflow fit

Built for catalog, SKU, packaging, QA, and ERP/product-data teams.  
Can become Android capture app + web dashboard + API integration.

## Short answer

The unique value is not “recognizing one product.” It is helping teams review visually similar products, duplicates, variants, and packaging mistakes across a catalog.

# Ecommerce Integration Opportunity

Shopify / WordPress product creation + AI-generated descriptions

## Where it connects

Shopify stores.  
WordPress / WooCommerce stores.  
ERP or product catalog systems.  
Internal admin dashboards.

## New product workflow

Upload product image.  
Check for visually similar existing products.  
Prevent duplicate catalog entry.  
Generate draft description for review.

## OpenAI API value

Generate product title ideas.  
Draft product descriptions.  
Suggest category/tags.  
Create SEO-friendly copy for ecommerce.

## Human review required

AI-generated descriptions should be reviewed by the business.  
The system should not invent unsupported ingredients, claims, certifications, or nutrition facts.

## Commercial benefit

Faster product onboarding.  
Cleaner catalog data.  
Fewer duplicate listings.  
Better product pages for online stores.

## Best pitch

“The same visual search can become an ecommerce catalog assistant: check duplicates first, then draft product descriptions for Shopify or WordPress.”

# Customer Questions Cheat Sheet

Use these to qualify Product Similarity leads quickly

## Problem discovery

- Do you manage many SKUs or product images?
- Do you have duplicate or visually similar packaging problems?
- Do your team's compare old vs new packaging versions?
- Do wrong product images appear in catalog, ecommerce, ERP, or inventory systems?
- Do you need to verify images against barcode or SKU records?

## Workflow and next step

- Do you use ERP, ecommerce, WMS, Shopify, WordPress/WooCommerce, or inventory software?
- Who owns product catalog quality or SKU data?
- Would mobile capture/search help QA, warehouse, sales, or field staff?
- Would a web dashboard help admins review duplicates and export reports?
- If we tested 100-500 product images, what should it detect first?

**Close: “Would it make sense to test this with a small sample of your product catalog after the event?”**